

Strong edge colouring conjecture

OPG-46475	L1 - Tractable	Partiellement résolu	Graph Theory
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Énoncé original (anglais, reproduction mot pour mot)

A strong edge-colouring of a graph G is a edge-colouring in which every colour class is an induced matching; that is, any two vertices belonging to distinct edges with the same colour are not adjacent. The strong chromatic index $\chi'(G)$ is the minimum number of colours in a strong edge-colouring of G .

Conjecture $\chi'(G) \leq \frac{5\Delta^2}{4}, \text{ if } \Delta \text{ is even,}$
 $\chi'(G) \leq \frac{5\Delta^2 - 2\Delta + 1}{4}, \text{ if } \Delta \text{ is odd.}$

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